Nomadology's two sciences

Deleuze and Guattari distinguish two kinds of science, or two conceptions of science, that share "a single field of interaction" (367). The **royal** science of the State reproduces, deduces, and extracts constants. In contrast, **minor** or nomad science follows a material and poses problems that it cannot solve on its own. The middle column, which is the reason for this sheet, records how one passes into the other, whether by translation, appropriation, repression, or the cutting loose of what was taken. It is the companion to *The five machinic processes*, which gives this whole matter a single row, "How is knowledge made?"

The borderline: translation, appropriation, repression, cutting loose · Dashed: a proposed extension, not a claim of ATP · R Finding from the record

Question	Royal science THE STATE'S SCIENCE	The borderline WHERE ONE PASSES INTO THE OTHER	Minor science NOMAD, AMBULANT, VAGUE
What is its model?	A theory of **solids**, treating fluids as a special case, and a model of the stable, the eternal, the identical, the constant (Euclid, Plato's *Timaeus*) 361	In the *Timaeus*, Plato raises becoming as a model "only in order to exclude it and conjure it away in the name of royal science" 361	A **hydraulic** model, in which "flux is reality itself," a model of becoming and heterogeneity, with the clinamen as the smallest deviation of an atom (Democritus, Lucretius, cf. Serres) 361
In what space does it work?	**Striated** or metric space, "counted in order to be occupied," the homogeneous space of gravity, "the space of pillars" 362, 370	Smooth space is "always translatable, and necessarily translated," by laying a Euclidean space over each of its points, "the triumph of the logos or the law over the nomos" 373	**Smooth** space, topological instead of metric, "occupied without being counted," a space of contact that can "be explored only by legwork" 362, 371
At what speed, to what rhythm?	**Gravitas**: laminar movement from point to point, and a "measured, cadenced" rhythm, like a river between its banks 363-64, 371	Their opposition is qualitative, a condition "that regulates both the separation and the mixing of the two models" 371	**Celeritas**: the movement that deviates and becomes vortical, and "a rhythm without measure," like the upswell of a flood 364, 371
How does it treat a figure?	**Theorematic**: from an essence to its properties, from a genus to its species, with the circle as "an organic, ideal, fixed essence" 362, 367	Royal science reduces "the 'problem-element'" and subordinates it to "the 'theorem-element'" 362	**Problematic**: figures as events (no square without a quadrature), and roundness as a vague essence, "anexact yet rigorous" (cf. Husserl) 362, 367
What does an equation do?	It extracts constants from variables, "an invariable form for variables," which Deleuze and Guattari call the **compars** 369 R4	Differential calculus was admitted as "a well-founded fiction" once its "dynamic, nomadic notions" had been eliminated (Carnot) 363	It puts variables into continuous variation ("adequations, inequations, differential equations"), the **dispars**, with equations generated by the material "as 'forces of thrust'" (the Gothic vault) 364-65, 369
How does it proceed?	**Reproducing**: iteration, deduction, induction, from a fixed point of view, "watching the flow from the bank" 372	"The question is not whether such a translation is conceptually legitimate - it is - but what intuition gets lost in it" 409	**Following**: itineration across a field where singularities are "scattered like so many 'accidents' (problems)" (the smith, the Gothic journeyman) 372, 409 R2
How do matter and form relate?	**Hylomorphism**: a form that organizes a matter prepared for it, a schema derived from "a society divided into governors and governed" 369	Between form and matter lies "a zone of medium and intermediary dimension" that the model cannot see (cf. Simondon) 408-09	**Material-forces**: matter "laden with singularities" and expression with "pertinent traits," so that one addresses "a materiality possessing a nomos" (splitting wood along its grain) 369, 408 R3
What plane does it draw?	The architect's metric plane, "on paper and off site," a plane of organization, with stone cut by templates as a model "for reproduction" 365, 368	Bernard de Clairvaux gave up founding Gothic building on Euclid as too "difficult" and turned to "an operative, Archimedean geometry" 364	The journeyman's ground-level plane, which is the plane of consistency, with stone cut by squaring, so that "the cutting line propels the equation" (Garin de Troyes) 364, 368
What division of labor does it assume?	The split between intellectual and manual labor, "modeled upon the difference between 'governors' and 'governed,'" with labor dequalified 368	The State represses nomad science not for being inexact but "because they imply a division of labor opposed to the norms of the State" 368	The itinerant bodies of masons, carpenters, smiths and the like (the compagnonnages), whose powers are "mobility and the strike," and the artisan who must also prospect 368, 409
How autonomous is it?	Autonomous, with "a metric power" that isolates its operations from intuition as "true intrinsic concepts, or 'categories'" 373-74 R1	The ambulant sciences invent problems whose "scientific solution depends ... on royal science" (cf. Bergson's intuition and intelligence) 374	Not autonomous: its solutions are "tied to a whole set of collective, nonscientific activities," and "problematics is still its only mode" 373-74
Who practices it, and what happens to them?	"The men of State of science," the great axiomaticians, "heirs of the theorematic conception of geometry" 461	Savants "between a rock and a hard place," above all the military engineer, used by State science "only after restraining or disciplining them" (Vauban, Monge, Carnot) 362-63	Archimedes "vanquished by the Roman State," Desargues "condemned by the parlement of Paris," the Gothic guilds banned, Perronet's bridges 362, 365 R5
What is its exemplary science?	Gravity, "the law of all laws," and geometry, so that "even chemistry became a royal science only by virtue of ... the notion of weight" 370	Each new field of science "proved irreducible to the field of attraction ... although not contradictory" to it, lodging itself in "an excess" 370	Metallurgy, "minor science in person, 'vague' science or the phenomenology of matter," and the smith who follows the ore 411-12
What does it do with problems?	It answers them with an **axiomatic**, "a stopping point, a reordering" that keeps decoded flows from "escaping in all directions" 461	Intuitionism opposed axiomatics "in the name of a calculus of problems," an abstract machine "working in the undecidable and the fugitive" 461	It poses them, as a constructivism "operating by the determination of the conditions of the problem and by transversal links between problems" 473
Where is it now?	*Machine learning predicts from data without a model of the process (cf. Breiman's "two cultures"), which makes it perhaps a royal science that has learned to follow*	*A measure kept in common (the sealings of Sabi Abyad, the Indus weights) against a measure owned by one party and applied to others (the platform's reputation score)*	*Knowledge kept by restriction and evasion, as in the Jamaican Maroons' "Maroon Science" (Bilby) or Saramaka history told in fragments (Price)*
How do the two meet?	"Two formally different conceptions of science, and, ontologically, a single field of interaction in which royal science continually appropriates the contents of vague or nomad science while nomad science continually cuts the contents of royal science loose. At the limit, all that counts is the constantly shifting borderline" 367		

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Working with another field

A nomadology works with other fields, here anthropology, archaeology, and history, without taking on their frame along with their findings. Each row below changes a question that those fields often ask from the side of the State into one that a nomadology can pursue, with the pages that ground it and a case from the specialist literature.

Instead of asking	A nomadology asks	ATP	A case
CAPTURE COMES SECOND			
How did the State originate?	**What does this capture make, and then claim to have found already there?**	427, 446-48	Will on Corinth: money came from taxation, not trade (442)
Did storage cause inequality?	**What capture does this store reveal?**	440-44	Ingold: a stock is rights converging "upon a single interest"
COEXISTENCE			
Which came first?	**What is warded off and anticipated here, and at what threshold would it take on consistency?**	431-34	Lévi-Strauss: one village, two potentials, segmentary or hierarchized (433)
Is this a State society or not?	**Which processes are at work, and which one sets the tone?**	435-37	Green: Indus cities with writing and weights but no palaces
THE OUTSIDE			
Who lived free of the State?	**Where does the border with this State run, and where does it give way?**	359-60	Casimir: cultivators who stopped being the plantation's workers without leaving it
SPACE AND NUMBER			
How mobile were they?	**What did they occupy without dividing or counting it?**	380-81, 410	Humphrey and Sneath prefer "mobile pastoralism," yet movement is not what defines the nomad (381)
How was the population counted?	**Does this count compose a body that moves, or record a resource?**	387-92	Sabi Abyad: sealings from stores shared "probably along egalitarian lines" (Frangipane)
How many escaped?	**What in this escape cannot be counted as a set?**	469-70	Price: Saramaka history kept restricted and in fragments
MINOR SCIENCE			
What do the data show?	**What was set aside as noise so that the data could be compared?**	373, 409	Moreno García and Richardson: royal measure without a royal science
What constants can we extract?	**What did this practice follow, and can our method follow it too?**	372	Carney and Rosomoff: maroon smithies on a 1769 map
Was this knowledge accurate?	**What kind of rigor does it have?**	367	Bilby: "Maroon Science," learned through "testing and evasion"
What form was imposed?	**What tendencies in the material were followed?**	408-09	Simondon: splitting wood along its fibers
Who designed this?	**What division of labor does this knowledge assume?**	368-69	The Gothic journeymen against the architect's plane on paper (368)
PROBLEMS AND AXIOMS			
What is the solution?	**Who gets to pose the problem, and what remains undecidable once it is posed?**	461, 471-73	Hudson: each royal debt cancellation as an axiom added to keep an order running
Was this resistance or destruction?	**Does this line create new relations, or does it only destroy?**	422-23	Munkh-Erdene: the warband of a warlord as the "nomadic origin of the state"

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Notes and sources

The markers R1-R5 on the front point to findings from anthropology, archaeology, and history that complicate a cell. Below them is a key to every scholar and place named on the sheet.

Findings from the record

R1 How autonomous is it? · Royal science — In fact, in ancient Egypt and Babylonia, fields were measured by a dozen kinds of officer with no fixed competence, which is to say royal measure without a royal science (Moreno García and Richardson).

R2 How does it proceed? · Minor science — Yet following can also be trained into the State's service. Tracking, which Louis Liebenberg presents as the origin of science among Kalahari hunters, is also the craft of the Shadow Wolves, a United States border unit of Native American trackers.

R3 How do matter and form relate? · Minor science — A 1769 militia map of a runaway settlement in Minas Gerais, Brazil, marks its smithies and gardens beside its hidden defenses and "holes for escape" (Carney and Rosomoff, 85).

R4 What does an equation do? · Royal science — Indeed, the Indus cities had standard weights, seals, and writing but no palaces or royal tombs (Green), so a fixed measure need not belong to a ruler.

R5 Who practices it, and what happens to them? · Minor science — Minor science still has practitioners. The Jamaican Maroons learn their "Maroon Science" through "testing and evasion" (Bilby), and the Saramaka of Suriname keep the history of their war against the planters restricted and in fragments (Price).

Who and what is named

Archimedes, Democritus, Lucretius
Ancient sources of minor science: Archimedes' geometry of curves and the atomism of Democritus and Lucretius, as Michel Serres treats them (361-62).

Bergson, Henri
The relation between the two sciences is "somewhat like intuition and intelligence in Bergson," where only intelligence can solve formally the problems that intuition poses (374).

Bernard de Clairvaux, Garin de Troyes
The twelfth-century abbot and, by legend, his monk-mason, whose "operative logic of movement" stands for the geometry of the Gothic journeymen (364).

Bilby, Kenneth
For the anthropologist Kenneth Bilby, in *True-Born Maroons* (2005), the Jamaican Maroons keep a knowledge they call "Maroon Science," learned through "testing and evasion."

Breiman, Leo
In "Statistical Modeling: The Two Cultures" (2001), the statistician Leo Breiman contrasts models of how data are generated with algorithms that predict without such a model.

Carney, Judith, and Richard Rosomoff
In *In the Shadow of Slavery* (2009), the geographers Judith Carney and Richard Rosomoff reproduce a 1769 militia map of a runaway settlement in Minas Gerais, Brazil, marking its smithies, gardens, hidden defenses, and "holes for escape" (85).

Casimir, Jean
The Haitian historian Jean Casimir shows in *The Haitians* (2020) how cultivators after the revolution built a "counter-plantation" of family farms and, "without moving from the plantations," stopped being the plantation's workers (142), a book Protevi also cites (RoV 104).

Desargues, Monge, Poncelet, Carnot, Vauban, Perronet
Engineers and mathematicians whom Deleuze and Guattari place on the borderline, from Vauban's fortifications and Desargues's perspective to Perronet's bridges (363-65).

Frangipane, Marcella
The archaeologist Marcella Frangipane describes the clay sealings of the "burnt village" at Sabi Abyad in northern Syria, some eight thousand years old, as records of stores shared "probably along egalitarian lines" (2018).

Green, Adam
In "Killing the Priest-King" (2021), the archaeologist Adam Green describes Indus cities (about 2600-1900 BCE) with writing, standard weights, seals, and metalwork, yet no palaces or royal tombs.

Hudson, Michael
The economic historian Michael Hudson reconstructs the royal cancellations of agrarian debt in the ancient Near East, the "clean slates" (2002).

Humphrey, Caroline, and David Sneath
Writing on Mongolia and Inner Asia in *The End of Nomadism?* (1999), the anthropologists Caroline Humphrey and David Sneath call nomadism "a category imagined by outsiders" and prefer "mobile pastoralism" (1).

Husserl, Edmund
Nomad science takes its anexact rigor from Edmund Husserl's "vague essences," roundness as opposed to the circle (367).

Indus, Sumer, Uruk
The Indus civilization flourished in present-day Pakistan and northwest India (about 2600-1900 BCE). Sumer names the city-states of southern Mesopotamia (from about 2900 BCE), while the period of Uruk, a Mesopotamian city (about 4000-3100 BCE), gives the earliest writing, kept as accounts.

Ingold, Tim
For the anthropologist Tim Ingold, setting food aside is not yet a stock, since a stock is rights converging "upon a single interest" (1983, 553).

Lévi-Strauss, Claude
Claude Lévi-Strauss showed that the same village can present itself as segmentary and egalitarian or as encompassing and hierarchized, two potentials that Deleuze and Guattari treat as anticipations of town and State (433).

Liebenberg, Louis, and the Shadow Wolves
In *The Art of Tracking: The Origin of Science* (1990), Louis Liebenberg studies Kalahari trackers. The Shadow Wolves are a unit of Native American trackers working for United States immigration enforcement on the Tohono O'odham Nation in Arizona.

Moreno García, Juan Carlos, and Seth Richardson
In *Monarchies and the Organization of Power* (2025), the historians Juan Carlos Moreno García and Seth Richardson find Egyptian and Babylonian kings co-opting local powers they could not erase (6), with fields measured by many kinds of officer (16).

Munkh-Erdene, Lhamsuren
In *The Nomadic Leviathan* (2023), the Mongolian historian Lhamsuren Munkh-Erdene argues "for the nomadic origin of the state" in the warband of a charismatic warlord (1, 25).

Price, Richard
In *First-Time* (1983), the anthropologist Richard Price records how the Saramaka of Suriname keep the history of their ancestors' war against the planters, restricted and told in fragments.

Querrien, Anne
The CERFI sociologist Anne Querrien, whose work on State engineering gives Deleuze and Guattari the Gothic cathedrals and the bridges of the Ponts et Chaussées (364-66).

Serres, Michel
In *The Birth of Physics* (1977), the philosopher Michel Serres finds a hydraulic physics in Lucretius, and with it the two sciences, "a general theory of routes and paths, and a global theory of waves" (361, 372).

Simondon, Gilbert
Following matter, as opposed to imposing form, draws on the philosopher Gilbert Simondon's critique of the form-matter, or hylomorphic, model (408).

Will, Édouard
The historian of archaic Greece Édouard Will, whose account of Corinth makes money derive from taxation (442).

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